



Deploying My portfolio Website on AWS with Terraform

Nathan Mersha Degineh, Cloud Programming Phase 2
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What i am trying to solve

High Availability

My Portfolio site must be up
even if one of the servers
went down

Low Latency

Regardless who visits the
site, it should be fast

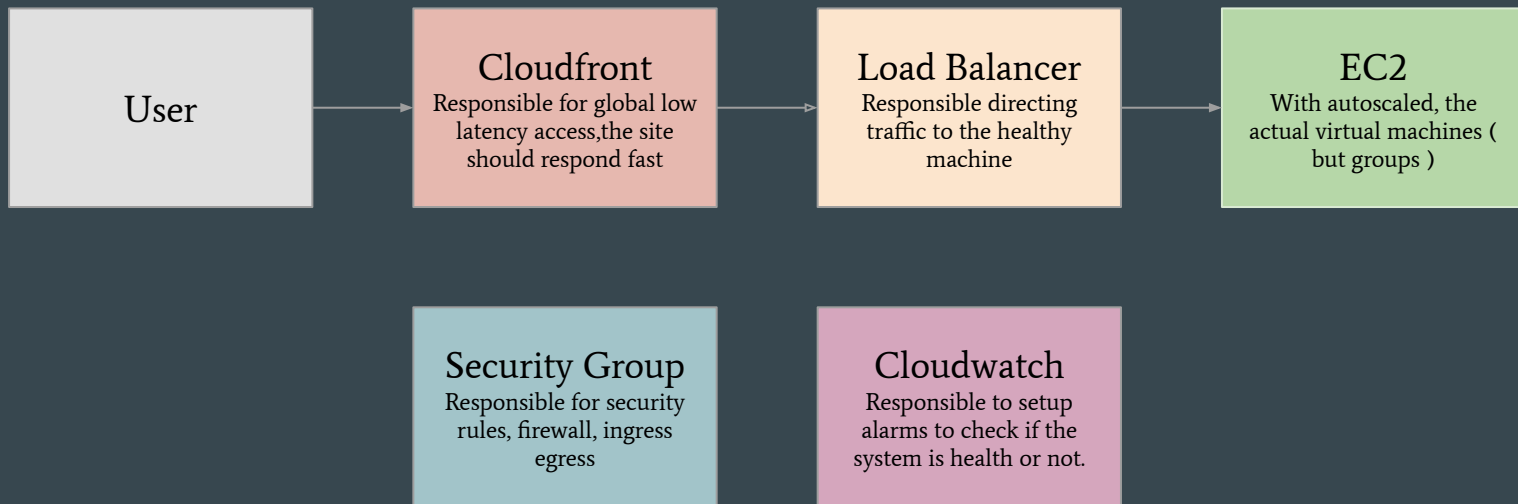
Scalable

The backend scales up when
the cpu is high and scales
down when traffic drops

<https://medium.com/@christianashedrack/building-a-highly-available-aws-infrast-structure-with-terraform-a-beginners-guide-4ab902cd2e33>

The Design

There are 5 aws components that work together for the portfolio site



CloudFront

What does it do?

- Global CDN with 400 edge location
- It caches content close to the user
- Free ssl certificate for my https site
- From Http to Https

Why I choose it?

- It's fast (no latency for the global users)
- This is accessed instead of the loadbalancer from the internet
- No cost for the ssl
- No direct load on my ec2s

Load Balancer

What does it do?

- Distributes incoming traffic across multiple ec2's
- Does health check for each ec2's
- Routes to the healthy ec2's
- Listens on port 80 and forwards to the right group.

Why I choose it?

- Highly available,
- It kills the unhealthy ec2's
- Available on multiple zones
- Greats work with cloudfront and the auto scaler

Auto Scaling Group

Configuration

- Minimum 2 instances always running
- Maximum up to 5 instances
- Ubuntu 22, with t3a.nano
- Installs Nginx, and serves simple html portfolio
- Health check enabled

How scaling happens!

- Add 1 ec2 when cpu > 75%
- Remove 1 ec2 when cpu < 25%
- Gets triggered by cloudwatch alarms

Security Groups

Load Balancer Security Group

- Inbound from port 80, (only accepts request from cloudfront)
- Use cloudfront prefix lists to validate where request is coming from
- No direct access directly to the load balancer from the internet
- There is no block for outbound requests

EC2 Security Group

- Inbound from port 80 (Accepts request only from the load balancer)
- Ec2 is completely not reachable from the internet
- User > CloudFront > LB > Ec2
- There is no block for outbound requests

Why Terraform?

Instead of clicking through the aws dashboard, everything is declared in terraform code, which means anyone can replicate

Project File Structure

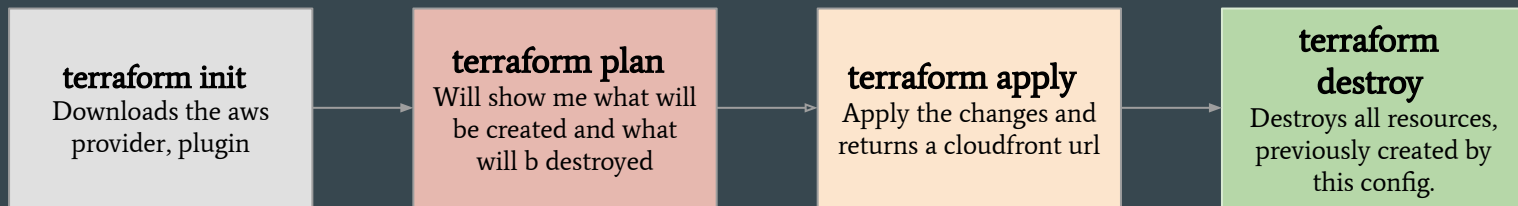
- [main.tf](#) : all AWS resources are defined here
- [variables.tf](#) : variables defined here
- [output.tf](#) : the cloudfront url can be found here as an output.

Resources Defined In Terraform

- 2 security groups (for the load balancer and for the ec2's)
- Load balancer + target group + listener
- Ec2 launch template + Autoscaling group
- 2 Cloudwatch Alarms and 2 Scaling policies (up and down)
- Cloudfront

How to deploy it?

The output from [output.tf](#) gives us https url from cloudfront, if we destroy and recreate everything, then the url will change, not associated with custom domain name



Summary

All 3 requirement are fulfilled, the portfolio site is :

High Availability

Low Latency

Auto Scaling

AWS
CloudFront

AWS
LoadBalancer

AWS
AutoScaller

AWS
CloudWatch

AWS
EC2

AWS
Security Groups

OutPut

GitHub

https://github.com/nathan-mersha/Cloud_Programming.git

CloudFront

(will delete it after the grade, because the setup is expensive, around 13\$ a month)

<https://d2k8aexqt4gygj.cloudfront.net>

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Reference

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